Armaan Shah

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EXPERIENCE

LogistieX | Software Engineer (Intern to Full-time) Jan 2026 - Present

- Designed an autoscaling asynchronous worker architecture on Cloud Run serving AI product-image generation for 100+ e-commerce sellers, handling concurrency through a Cloud Tasks queue and worker router.
- Implemented a retrieval-augmented caching layer that surfaces prior schema mappings, transformation rules and resolved edge cases at runtime, reaching a roughly 60% hit rate and cutting redundant LLM regeneration across distributed processing jobs.
- Built a pure-Python Excel formula engine (tokenizer, parser and evaluator covering 15 functions) that evaluates workbook formulas, conditional formatting and dropdown rules with no Excel automation, so the pipeline runs on Linux where Excel tooling is unavailable.
- Built the upload and sync path for human review of AI-generated schemas: an API route and service that diff a reviewer's edited data against the original and apply only the changes back to the JSON.

Deloitte Touche Tohmatsu India LLP | Software Engineering Intern May 2025 - Aug 2025

- Built key functionality for the payment module of a top-3 corporate banking platform, engineering the end-to-end data flow from a secure Express and SQL backend to a Redux Saga frontend.
- Strengthened API security with a hybrid AES/RSA payload encryption protocol, and ran the session token refresh timer in a Web Worker to keep its timing consistent (±7ms).
- Engineered a micro-frontend architecture by post-processing React's bundled runtime, letting Next.js apps embed inside each other while maintaining independent state, and enforcing strict CSP and server-side rendering.

Mphasis AI | Machine Learning Applications - NLP Intern Jan 2025 - Apr 2025

- Developed (React, Flask, MongoDB) and deployed (AWS EC2) a full-stack HITL platform with auth, dashboards, and logging.
- Raised throughput by letting multiple reviewers validate in parallel, handling conflicting edits via a mutex and queue, and added lookahead caching to speed up database queries.
- Built a multi-stage data fidelity framework, using trained NER and QA models, that automatically flagged 83% of errors in LLM-extracted structured data.

DocsTribe AI | Software Engineering Intern May 2024 - Aug 2024

- Built full-stack features for two products from client requirements, one on a modern stack (React.js, Django) and one on a legacy system (PHP, Flask), processing 10k+ patients per day across hospitals.
- Cut worst-case server response times from 30-40 seconds to about 5 seconds by re-engineering the database query layer with MongoDB aggregation pipelines and server-side pagination.
- Reduced operational costs by 60% by executing a cloud migration from GCP to Microsoft Azure, consolidating servers behind a single proxy and automating TLS certificate management with Caddy. Infrastructure configured using Terraform.

PUBLICATION

Validating FKG.in: Soundness Assessment in LLM-Augmented Indian Food Knowledge. Gupta, **Shah**, Dey, Das, Jain. Accepted at JOWO/IFOW 2026. arXiv:2608.29249.

PROJECTS

QToday | AI-Native Education Platform

- Designed and implemented the data model and architecture of an edtech platform (FastAPI, PostgreSQL, React), and built an LLM agent that answers questions scoped to user permissions, with every number computed in SQL, reaching 98% accuracy on about 250 test questions with Gemma 4 12B.

Activity Matching App

- Built a cross-platform activity-matching app (.NET, React Native) with a Google Maps API integration and a two-layer recommendation system, using cached preference vectors for instant session startup and a multi-tier proximity cache that cut slow requests by about 40%.

iosc | Windows-Native iOS Build Toolchain [GitHub]

- Built iosc, a terminal toolchain that compiles, signs and installs full iOS apps from Windows.
- Implemented Apple's undocumented binary formats (asset catalogs, interface files, Swift macro plugins) from scratch by decoding real samples, along with the code-signing chain and on-device install.

Microfluidic Simulator | Undergraduate Research Thesis

- Built a high-performance microfluidics simulator in C++ (AVX2, Eigen) for my undergraduate research thesis, a

reduced-order laminar-flow model matching OpenFOAM CFD results within 3% while running about 270,000 times faster, exposed to Python through PyBind11 for reinforcement learning training.

SKILLS

Languages: Python, JavaScript, TypeScript, C++, C#, SQL
Backend: FastAPI, Flask, Django, Node.js, Express, ASP.NET Core
Frontend: React, Next.js, React Native, Redux Saga
Data: PostgreSQL, MongoDB, MS SQL Server
Cloud & DevOps: GCP, Cloud Run, Azure, AWS, Docker, Terraform, CI/CD
AI/LLM: LangGraph, LLM APIs, RAG, NER

EDUCATION

Ashoka University, Sonepat | BSc Computer Science | CGPA 3.61/4.0 (Major) 2022 - 2026

- ISC: 98% | ICSE: 97.5%
- Teaching Assistant, Discrete Mathematics (Monsoon 2025): mentored and graded 50+ students.
- Head of Development, CS Society: led 15 students building a client-developer matching portal, a university competition site (Next.js), and internal digitisation projects for university administration.